Darcy First Proof

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October 12, 2026

1 Problem Statement

Let $D \subseteq \mathbb{R}^d$, $d \geq 2$, denote the d-dimensional Euclidean unit ball. Given a function $f \in C^\alpha(D)$ with Hölder-Zygmund smoothness $\alpha > 1$ on D and let $u = u_f$ denote the unique solution of the following divergence form PDE (Darcy's problem)

$$\operatorname{div}(f\nabla u) = 1 \text{ on } D, \quad u|_{\partial D} = 0.$$

Consider the non-linear statistical inverse problem of reconstructing f from noisy measurements of u corrupted by Gaussian white noise

$$Y_n = u_f + \frac{1}{\sqrt{n}}\mathbb{W}.$$

For $r \in (0,1)$ consider the following class of conductivities

$$f \in \mathcal{F}(\alpha, r) = \{h \in C^\alpha(D),\ \|h - \mathbf{1}\|_{C^\alpha(D)} \leq r\},$$

where $\mathbf{1}$ denotes a constant function with value 1. For small (but fixed) $r \in (0,1)$, what is the minimax rate as $n \to \infty$ for estimating f in the L^2-mean squared error

$$\inf_{\hat{f}_N} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0}\left[\|\hat{f}_n - f_0\|^2_{L^2(D)}\right],$$

up to logarithmic factors?

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